Question #1 of 111

Question ID: 1473752

Which of the following statements regarding an option's price is CORRECT? An option's price is:

A) an increasing function of the underlying asset's volatility. ✓

B) a decreasing function of the underlying asset's volatility. ✗

C) a decreasing function of the underlying asset's volatility when it has a long time remaining until expiration and an increasing function of its volatility if the option is close to expiration. ✗

Explanation

Since an option has limited risk but significant upside potential, its value always increases when the volatility of the underlying asset increases.

(Module 31.7, LOS 31.k)

Question #2 of 111

Question ID: 1473755

Two call options have the same delta but option A has a higher gamma than option B. When the price of the underlying asset increases, the number of option A calls necessary to hedge the price risk in 100 shares of stock, compared to the number of option B calls, is a:

A) larger positive number. ✗

B) larger (negative) number. ✗

C) smaller (negative) number. ✓

Explanation

For call options larger gamma means that as the asset price increases, the delta of option A increases more than the delta of option B. Since the number of calls to hedge is (–1/delta)x(number of shares), the number of calls necessary for the hedge is a smaller (negative) number for option A than for option B.

(Module 31.7, LOS 31.l)

Question #3 of 111

Question ID: 1473748

How is the gamma of an option defined? Gamma is the change in the:

A) delta as the price of the underlying security changes.

B) vega as the option price changes.

C) option price as the underlying security changes.

Explanation

Gamma is the rate of change in delta. It measures how fast the price sensitivity changes as the underlying asset price changes.

(Module 31.7, LOS 31.k)

Question #4 of 111

Question ID: 1473685

Referring to put-call parity, which one of the following alternatives would allow you to create a synthetic stock position?

A) Buy a European call option; buy a European put option; invest the present value of the exercise price in a riskless pure-discount bond.

B) Buy a European call option; short a European put option; invest the present value of the exercise price in a riskless pure-discount bond.

C) Sell a European call option; buy a European put option; short the present value of the exercise price worth of a riskless pure-discount bond.

Explanation

According to put-call parity we can write a stock position as: $S_0 = C_0 - P_0 + X/(1+R_f)^T$

We can then read off the right-hand side of the equation to create a synthetic position in the stock. We would need to buy the European call, sell the European put, and invest the present value of the exercise price in a riskless pure-discount bond.

(Module 31.1, LOS 31.a)

Question #5 of 111

Question ID: 1473743

The value of a put option is positively related to all of the following EXCEPT:

A) risk-free rate.

B) exercise price.

C) time to maturity.

Explanation

The value of a put option is negatively related to increases in the risk-free rate.

(Module 31.7, LOS 31.k)

Max Perrot, CFA, works for WWF, a mortgage banking company which originates residential mortgage loans. On a monthly basis, WWF issues agency mortgage-backed securities (MBS) backed by their loans. WWF sells the MBS in the open market soon after securitization, but retains the servicing rights to the loans. WWF currently owns the third largest mortgage servicing portfolio in the U.S. Perrot has recently been promoted to Senior Vice President of Asset and Liability Management for WWF. Perrot's new responsibilities encompass hedging WWF's newly created MBS prior to their sale, as well as managing the interest rate exposure on the servicing portfolio. Both types of assets are extremely sensitive to changes in interest rates, though not necessarily in the same manner.

Although WWF has retained all of the servicing rights of its loans in the past, they are not opposed to the selling of portions of the portfolio if market conditions are right. WWF's management wants Perrot in his new position to focus primarily on preserving the value of the servicing portfolio through hedging strategies that are cost effective to execute. Also, any hedge strategy used by Perrot must be extremely liquid in the event that a portion of the servicing portfolio is sold and the hedge needs to be unwound. The upper management of WWF anticipates a period of volatility in interest rates, and they have asked Perrot to project expected returns of a hedged position under a variety of interest rates scenarios.

Perrot's predecessor lacked experience in hedging with swaps and futures contracts, but he had used them periodically with lackluster results. Through his inaction, he had exposed the firm to significant asset and liability mismatch, which had increased dramatically over the past two years as both production and the servicing portfolio had grown. Perrot, on the other hand, had extensive experience with hedging with derivatives in his prior job. He is familiar with executing hedging strategies utilizing not only swap and futures, but also with options such as caps and floors. He decides that before he presents any potential hedging strategy to WWF's management, he would first like to bring them up to speed on the basic hedging concepts. He prepares a brief presentation on the relationships between interest rates and options, and outlines some basic hedging strategies. He anticipates many questions that may arise from his presentation, and prepares a handout in a question and answer format.

Question #6 - 9 of 111

Question ID: 1586311

Assume that a three-year semi-annually settled floor with a strike rate of 8% and a notional amount of $100 million is being analyzed. The reference rate is six-month London Interbank Offered Rate (LIBOR). Suppose that LIBOR for the next four semi-annual periods is as follows:

Period	LIBOR
1	7.5%
2	8.2%
3	8.1%
4	8.7%

What is the payoff for the floor for period 1?

A) $0.

B) $250,000.

C) $500,000.

Explanation

The payoff for each semi-annual period is computed as follows:

Payoff = notional amount × (floor rate − six-month LIBOR) / 2

so for period 1:

= $100 million × (8.0% − 7.5%) / 2 = $250,000

(Module 31.6, LOS 31.j)

Question #7 - 9 of 111

Question ID: 1586312

Which of the following *best* explains the difference between an interest rate put option and a put option on a fixed income security? The interest rate put option value:

A) decreases if interest rates increase just as the value of a put option on a fixed income security decreases.

B) decreases if interest rates increase while the value of a put option on a fixed income security increases if interest rates increase.

C) increases if interest rates increase just as the value of a put option on a fixed income security increases.

Explanation

An interest rate put option pays off the difference between the strike rate and the current interest rate if that difference is positive. So the value of the interest rate option will be high if interest rates decrease below the strike rate. In contrast, a put option on a fixed income security has a high value if interest rates increase because then the fixed-income security's price decreases below the value based strike price.

(Module 31.6, LOS 31.j)

Question #8 - 9 of 111

Question ID: 1586313

A LIBOR based floating rate bond combined with a LIBOR based collar (a short position in an interest rate cap and a long position in an interest rate floor both at the same strike rate) is equivalent to a:

A) pay-fixed swap position. ✗

B) fixed-rate bond. ✓

C) call option on a bond. ✗

Explanation

The effective rate above the cap strike and below the floor strike, when combined with the floating rate on a bond, is constant.

(Module 31.6, LOS 31.j)

Question #9 - 9 of 111

Question ID: 1586314

Which of the following is *most likely* a reason why dynamic riskless arbitrage is difficult in real markets?

A) Securities are subject to insider trading. ✗

B) Continuous rebalancing. ✓

C) Short sale constraints exist. ✗

Explanation

The continuous rebalancing required with dynamic riskless arbitrage is not practical. For one thing, it leads to significant transaction costs.

(Module 31.6, LOS 31.j)

You are interested in derivative products, particularly with a view to identifying arbitrage opportunities. You start with bond futures:

- The cheapest to deliver (CTD) bond underlying the T-bond futures contract maturing in five months is a 4.6% T-Bond currently priced at $1,002.33 (full price) per $1,000 par. The CTD paid its last coupon four months ago, and its conversion factor is 1.13. The risk free rate is 2.99%.

Peter Wang, one of your colleague, knew of your interest in derivative products advises you to consider interest rate options and swaptions. Wang makes the following comments:

Comment 1:	An investor having a long position in a call option on a bond has the same position as if he is long an interest rate floor.
Comment 2:	A borrower of a floating rate loan can create an interest rate collar by buying an interest rate cap and selling an interest rate floor and the cap sets the maximum interest rate payable by the borrower.
Comment 3:	A payer swaption is the right to enter into a specific swap at some date in the future as the fixed-rate payer. A payer swaption becomes more valuable if an equivalent swap at the market rate is higher than the strike rate.

Question #10 - 13 of 111

Question ID: 1473723

Which of the following is *closest* to the no-arbitrage price of the 5-month T-Bond futures contract?

A) $867.20. ✓

B) $877.47. ✗

C) $976.02. ✗

Explanation

The no-arbitrage price for T-Bond futures is given by the formula:

QFP = {(full price) × $(1 + R_f)^T$ – AI_T – FVC)(1 / CF)

The full price of the bond = clean price + accrued interest. Since the bond pays semi-annual coupons, and four months have passed since the last coupon, there are two months until the next coupon.

Accrued interest (AI) = (days since last coupon / days between coupons) × $ semiannual coupon. In this question we have not been told days but instead have months.

AI_T in the formulae represents the accrued interest at the maturity of the futures contract. Given the last coupon was 4 months ago the next coupon of $23 will be in two months' time. At the maturity of the futures contract in five months we will be 3 months through the coupon period, hence:

AI_T = (3 months / 6 months) × $23 = $11.5

The next coupon will be in two months' time (four months' ago, plus six months) and will equal $23. FVC in the above formula is this coupon compounded up to the futures maturity, three months later, so FVC = $23 × 1.02993 / 12 = $23.17.

Thus, QFP = [($1,002.33 × 1.02995 / 12) – $11.5 – $23.17] / 1.13 = $867.20

(Module 30.3, LOS 30.d)

Question #11 - 13 of 111

Question ID: 1473724

Comment 1 is *best* described as:

A) correct.

B) incorrect as long an interest rate floor should be short an interest rate floor.

C) incorrect as long an interest rate floor should be long an interest rate cap.

Explanation

For a long call option on a bond, when interest rates decrease, bond prices rise hence call value increases. Similarly, for an interest rate put, when interest rate decreases, the long put value increases.

(Module 31.6, LOS 31.j)

Question #12 - 13 of 111

Question ID: 1473725

Comment 2 is *best* described as:

A) correct.

B) incorrect as it should be buying an interest rate floor and selling an interest rate cap.

C) incorrect as it should be the floor that sets the maximum interest rate payable by the borrower.

Explanation

Interest rate cap pays when the reference rate exceeds the strike rate. As such, the borrower can use the payment from the interest rate cap to offset the higher interest payment on the floating rate loan. Hence the strike rate on the cap is the maximum interest rate that the borrower has to pay. By selling the floor, the premium received on the floor will help to offset some of the premium paid on the cap. In addition, the collar also hedged the interest rate exposure of the loan through the strike rate of the cap and floor.

(Module 31.6, LOS 31.j)

Question #13 - 13 of 111

Question ID: 1473726

Comment 3 can be *best* described as:

A) correct.

B) incorrect as it is describing a receiver swaption, not a payer swaption.

C) incorrect as a payer swaption is more valuable if an equivalent swap at the market rate is lower than the strike rate.

Explanation

Comment 3 is a correct description of a payer swaption.

(Module 31.6, LOS 31.j)

Question #14 of 111

Question ID: 1473753

Compared to the delta of a long position in a stock, the delta of an at-the-money call option on the stock is *most likely* to be:

A) the same.

B) less.

C) greater.

Explanation

The delta of an at-the-money call option is typically close to 0.5. The delta of a long position in the underlying stock is 1.0 by definition.

(Module 31.7, LOS 31.k)

Question #15 of 111

Question ID: 1473705

Which of the following is NOT one of the assumptions of the Black-Scholes-Merton option-pricing model?

A) The yield curve for risk-free assets is fixed over the term of the option.

B) There are no taxes and transactions costs are zero for options and arbitrage portfolios.

C) Early exercise is not allowed.

Explanation

The yield curve is assumed to be flat so that the risk-free rate of interest is known and *constant* over the term of the option. Having a fixed yield curve does not necessarily imply that the yield curve is flat. BSM assumptions include that markets are frictionless (no taxes, transactions costs) and that the options are European-style, meaning that early exercise is not allowed.

(Module 31.6, LOS 31.f)

Frank Potter, CFA, a financial adviser for Star Financial, LLC has been hired by John Williamson, a recently retired executive from Reston Industries. Over the years Williamson has accumulated $10 million worth of Reston stock and another $2 million in a cash savings account. Potter has a number of unconventional investment strategies for Williamson's portfolio; many of the strategies include the use of various equity derivatives.

Potter's first recommendation involves the use of a total return equity swap. Potter outlines the characteristics of the swap in Table 1. In addition to the equity swap, Potter explains to Williamson that there are numerous options available for him to obtain almost any risk return profile he might need. Potter suggest that Williamson consider options on both

Reston stock and the S&P 500. Potter collects the information needed to evaluate options for each security. These results are presented in Table 2.

Table 1: Specification of Equity Swap

Term	3 years
Notional principal	$10 million
Settlement frequency	Annual, commencing at end of year 1
Fairfax pays to broker	Total return on Reston Industries stock
Broker pays to Fairfax	Total return on S&P 500 Stock Index

Table 2: Option Characteristics

	Reston	S&P 500
Stock price	$50.00	$1,400.00
Strike price	$50.00	$1,400.00
Interest rate	6.00%	6.00%
Dividend yield	0.00%	0.00%
Time to expiration (years)	0.5	0.5
Volatility	40.00%	17.00%
Beta Coefficient	1.23	1
Correlation	0.4	

Table 3: Regular and Exotic Options (Option Values)

	Reston	S&P 500
European call	$6.31	$6.31
European put	$4.83	$4.83
American call	$6.28	$6.28
American put	$4.96	$4.96

Table 4: Reston Stock Option Sensitivities

	Delta
European call	0.5977
European put	–0.4023

American call	0.5973
American put	–0.4258

Table 5: S&P 500 Option Sensitivities

	Delta
European call	0.622
European put	–0.378
American call	0.621
American put	–0.441

Potter has also been asked to evaluate the interest rate risk of an intermediate size bank. The bank has a large floating rate liability of $100,000,000 on which it pays the MRR on a quarterly basis. Potter is concerned about the significant interest rate risk the bank incurs because of this liability: since most of the bank's assets are invested in fixed rate instruments there is a considerable duration mismatch. Some of the bank's assets are floating rate notes tied to MRR, however, the total par value of these securities is significantly less than the liability position.

Potter considers both swaps and interest rate options. The interest rate options are 2-year caps and floors with quarterly exercise dates. Potter wishes to hedge the entire liability.

Potter has obtained the prices for an at-the-money 6 month cap and floor with quarterly exercise. These are shown in Table 6.

Table 6: At-the-Money 0.5 year Cap and Floor Values

Price of at-the-money Cap	$133,377
Price of at-the-money Floor	$258,510

Question #16 - 19 of 111

Question ID: 1473757

Williamson would like to consider neutralizing his Reston equity position from changes in Reston's stock price. Using the information in Tables 3 and 4 how many standard Reston European options would have to be bought/sold in order to create a delta neutral portfolio?

A) Sell 370,300 call options.

B) Sell 497,141 put options.

C) Buy 497,141 put options.

Explanation

Number of put options = (Reston Portfolio Value / Stock $\text{Price}_{\text{Reston}}$) / −DeltaPut

Number of put options = ($10,000,000 / $50.00) / −0.4023 = −497,141 meaning buy 497,141 put options.

(Module 31.7, LOS 31.l)

Question #17 - 19 of 111

Question ID: 1473758

Williamson is very interested in the total return swap. He asks Potter how much it would cost to enter into this transaction. Which of the following is the *most likely* cost of the swap at inception?

A) $45,007.

B) $0.

C) $340,885.

Explanation

Swaps are priced so that their value at inception is zero.

(Module 30.8, LOS 30.g)

Question #18 - 19 of 111

Question ID: 1476157

Potter analyzes alternative hedging strategies to address the risk of the bank's large floating-rate liability. Which of the following is the *most appropriate* transaction to efficiently hedge the interest rate risk for the floating rate liability without sacrificing potential gains from interest rate decreases?

A) Sell an interest rate floor and buy an interest rate cap.

B) Sell an interest rate cap.

C) Buy an interest rate cap.

Explanation

Buying a cap, combined with a floating rate liability, limits the exposure to interest rate increases (i.e., no exposure to interest rate increases above strike rate). The floating rate borrower will still benefit from interest rate decreases.

(Module 31.6, LOS 31.j)

Question #19 - 19 of 111

Question ID: 1473760

Potter is now considering some of the bank's floating rate assets. Which of the following transactions is the *most appropriate* to minimize the interest rate risk of these assets without sacrificing upside gains?

A) Buy a floor.

B) Buy a cap.

C) Buy a collar.

Explanation

Buying a floor combined with a floating rate assets limits the exposure to interest rate decreases (i.e., no exposure to interest rate decreases below strike rate) while the floating rate holder is still able to benefit from interest rate increases. Ideally, Potter should consider matching the bank's asset position against the bank's liability position.

(Module 31.6, LOS 31.j)

Question #20 of 111

Question ID: 1473710

Regarding options on a stock without dividends, it is:

A) sometimes worthwhile to exercise calls early but not puts.

B) sometimes worthwhile to exercise puts early but not calls.

C) never worthwhile to exercise puts or calls early.

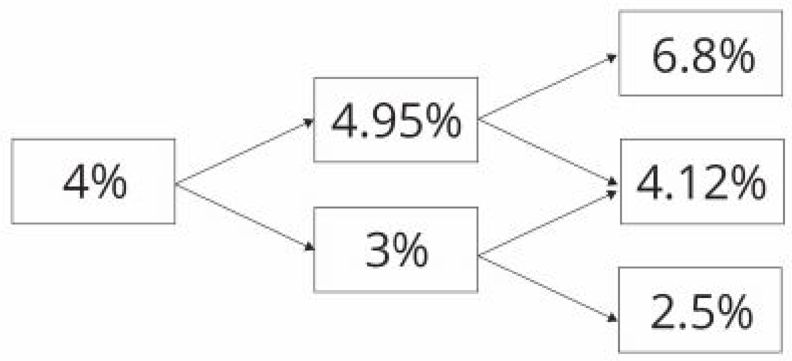
Explanation

After early exercise of a put, and in particular a deep in-the-money put, the sale proceeds can be invested at the risk-free rate and may earn interest worth more than the time value of the put option. The same is not true for call options: early exercise of call options on non-dividend-paying stock is never optimal.

(Module 31.6, LOS 31.i)

Lowell Wood is using the binomial option-pricing model to price interest rate options. She has obtained the following 2-year, annual rate tree (based on an assumed volatility of interest rates of 25%).

Exhibit 1



Wood has been asked help a colleague with the valuation of an interest rate put. The interest rate put option has 2 years to maturity and a strike price of 4.5% and is based on 360 day MRR. The option has a notional principal of $10m.

Wood has discovered that the Black model may be used to price options on interest rates by viewing the interest rate option as an option on a FRA. She is currently writing a research note for her team and makes the following three notes regarding the Black model:

Note 1:	"When using the Black model care needs to be taken to ensure that the payoff is discounted from the end of the borrowing and lending period (i.e., the maturity of the rate underlying the FRA), rather than the exercise date of the option."
Note 2:	"Given an interest rate option is an option on a FRA, call options will gain in value when interest rates rise and put options will fall in value."
Note 3:	"The accrual period needs to be factored in when valuing the option. This is because quoted rates are annual rates but in reality, the time between the FRA expiration and the maturity of borrowing and lending may not be one year. The accrual period can be viewed as a fraction of a year."

Wood asks for information about interest rate caps and floors. Newman makes the following comments:

Comment 1:	"A long FRA can be viewed as equivalent to a long interest rate call and a short interest rate put with the same strike and time to expiration."
Comment 2:	"Given a cap is a series of interest rate call options with identical strike prices and a floor is a series of interest rate put options also with identical strike prices, a short cap and long floor with identical strike prices would create a pay fixed receive floating interest rate swap."

Using the information about the interest rate put and the spot and forward rates in Exhibit 1, which of the following is *closest* to the value of the put? Assume that the option cash settle at time 2.

A) $44,250.

B) $64,250.

C) $84,250.

Explanation

Step 1:

At the expiry of the option at T_2 consider whether the option will be exercised in each of the forward scenarios.

Remember an interest rate put option allows the holder the right but not obligation to pay floating and receive fixed. The strike price (in this case 4.5%) is the fixed rate in an interest rate option. The put will be exercised in the forward rate is less than the fixed rate.

Forward rate 6.8% – option lapses

Forward rate 4.12% – option exercised

Forward rate 2.5% – option exercised

Step 2:

We now calculate the pay off on the option at the options expiry given each interest rate scenario. This assumes the payoff on the option is at T_2, which is technically incorrect as in reality the payoff is at the end of the borrowing and lending period (T_3) rather than the expiry of the option (T_2).

If we exercise the interest rate option we then enter pay floating receive fixed until the end of the borrowing and lending period.

Calculate the payoff on the put at T_2 in each scenario:

(interest rate received – interest rate paid) × days / 360 × notional principal

forward rate 6.8% – option lapses payoff zero

forward rate 4.12% = (0.045 – 0.0412) × 360 / 360 × $10m = $38,000

forward rate 2.5% = (0.045 – 0.025) × 360 / 360 × $10m = $200,000

Step 3:

Discount the payoffs back through the binomial interest rate tree to T_0. Note that in a binomial interest rate tree we always have a 50% chance of an up move and a 50% chance of a down move. Discount the probability weighted amounts from T_2 back to T_1 at the relevant forward rate.

Value at T_1 upper node:

[($0 + $38,000) ½] / 1.0495 = $18,103.86

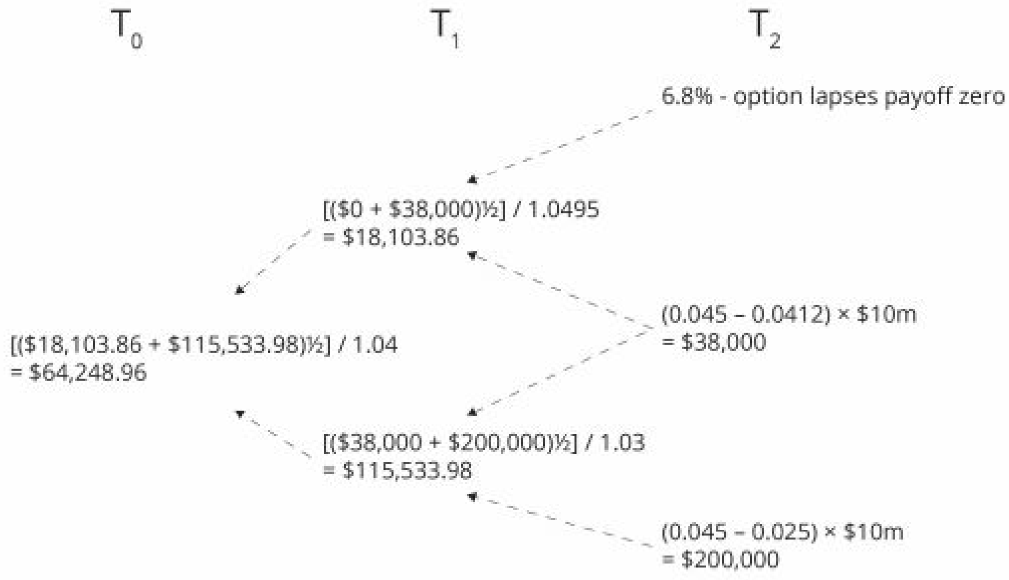
Value at T1 lower node:

[($38,000 + $200,000) ½] / 1.03 = $115,533.98

Discount the probability weighted amounts back from T1 to T0 at the 1 period spot rate to arrive that the price (premium) on the interest rate put.

[($18,103.86 + $115,533.98) ½] / 1.04 = $64,248.96

Diagrammatically:



(Module 31.5, LOS 31.e)

Question #22 - 24 of 111

Question ID: 1473714

How many of Wood's notes regarding the Black model used to value interest options are correct?

A) All three notes are correct.

B) Only two of the notes are correct.

C) Only one of the notes is correct.

Explanation

Note 1: Correct

The payoff is not at the expiration of the option and FRA. Once exercised into the FRA the payoff occurs at the end of the pay fixed receive floating period (FRA maturity).

Note 2: Correct

Long FRA represents pay fixed and receive floating from the expiry of the FRA until the end of the borrowing and lending period. Long FRAs will increase in value as interest rates rise as the forward rate on a new FRA with the same expiry and borrowing and lending period will be higher. Given the underlying in an interest rate option is a long FRA the call option will increase in value as the FRA gains value. As with all options the call will increase in value as the underlying increases.

Note 3: Correct

In the Black model used to value interest rate option the FRA rate and strike price are annualized rates. Given the time between FRA expiry and the end of borrowing and lending is unlikely to be a year the present value of the payoff will need to be unannualized.

(Module 31.6, LOS 31.j)

Question #23 - 24 of 111

Question ID: 1473715

Newman's Comment 1 is *best* described as:

A) correct. ✓

B) incorrect as to the strike price of the options. ✗

C) incorrect as to the equivalence to a long FRA. ✗

Explanation

Newman has replicated a long FRA position (pay fixed receive floating).

Long call, short put with same strike and expiry = Long FRA

Short call, long put with same strike and expiry = Short FRA

(Module 31.6, LOS 31.j)

Question #24 - 24 of 111

Question ID: 1473716

Newman's Comment 2 is *best* described as:

A) correct.

B) incorrect as buying a floor is not equivalent to buying interest rate put options.

C) incorrect as long floor, short cap would create a pay floating, receive fixed interest rate swap.

Explanation

Pay fixed, receive floating swap (payer swap) can be created by going long an interest rate cap and short an interest rate put.

A cap comprises of caplets where each caplet is in effect a call option on a FRA. A floor represents a series of floorlets where each floorlet is an interest rate put option.

Long floor and short cap with identical strike prices will create receiver swap (i.e., pay floating and receive fixed).

(Module 31.6, LOS 31.j)

Question #25 of 111

Question ID: 1473735

Mark Roberts anticipates utilizing a floating rate line of credit in 90 days to purchase $10 million of raw materials. To get protection against any increase in the expected MRR yield curve, Roberts should:

A) write a receiver swaption.

B) buy a payer swaption.

C) buy a receiver swaption.

Explanation

A payer swaption will give Roberts the right to pay a fixed rate below market if rates rise.

(Module 31.6, LOS 31.j)

Question #26 of 111

Question ID: 1473734

A payer swaption gives its holder:

A) the right to enter a swap in the future as the floating-rate payer.

B) an obligation to enter a swap in the future as the fixed-rate payer.

C) the right to enter a swap in the future as the fixed-rate payer.

Explanation

A payer swaption give its holder the right to enter a swap in the future as the fixed-rate payer.

(Module 31.6, LOS 31.j)

Question #27 of 111

Question ID: 1473703

Which of the following is NOT one of the assumptions of the Black-Scholes-Merton (BSM) option-pricing model?

A) There are no transaction costs, regulatory constraints, or taxes.

B) The options valued are European style (early exercise is not allowed).

C) The yield on the underlying has a known and constant volatility.

Explanation

Assumptions of BSM include:

- The volatility of the return on the underlying is known and constant.
- If the underlying instrument pays a yield, it is expressed as a continuous known and constant yield at an annualized rate.

(Module 31.6, LOS 31.f)

Question #28 of 111

Question ID: 1473687

Referring to put-call parity, which one of the following alternatives would allow you to create a synthetic European call option?

A) Sell the stock; buy a European put option on the same stock with the same exercise price and the same maturity; invest an amount equal to the present value of the exercise price in a pure-discount riskless bond.

B) Buy the stock; sell a European put option on the same stock with the same exercise price and the same maturity; short an amount equal to the present value of the exercise price worth of a pure-discount riskless bond.

C) Buy the stock; buy a European put option on the same stock with the same exercise price and the same maturity; short an amount equal to the present value of the exercise price worth of a pure-discount riskless bond.

Explanation

According to put-call parity we can write a European call as: $C_0 = P_0 + S_0 - X/(1+R_f)^T$

We can then read off the right-hand side of the equation to create a synthetic position in the call. We would need to buy the European put, buy the stock, and short or issue a riskless pure-discount bond equal in value to the present value of the exercise price.

(Module 31.1, LOS 31.a)

Question #29 of 111

Question ID: 1473750

The value of a put option will be higher if, all else equal, the:

A) exercise price is lower.

B) underlying asset has positive cash flows.

C) underlying asset has less volatility.

Explanation

Positive cash flows in the form of dividends will lower the price of the stock making it closer to being "in the money" which increases the value of the option as the stock price gets closer to the strike price.

(Module 31.7, LOS 31.k)

Question #30 of 111

Question ID: 1473761

When an option's gamma is higher:

A) a delta hedge will be more effective.

B) delta will be higher.

C) a delta hedge will perform more poorly over time.

Explanation

Gamma measures the *rate of change* of delta (a high gamma could mean that delta will be higher or lower) as the asset price changes and, graphically, is the curvature of the option price as a function of the stock price. Delta measures the slope of the function at a point. The greater gamma is (the more delta changes as the asset price changes), the worse a delta hedge will perform over time.

(Module 31.7, LOS 31.m)

Question #31 of 111

Question ID: 1473749

Which of the following is the *best* approximation of the gamma of an option if its delta is equal to 0.6 when the price of the underlying security is 100 and 0.7 when the price of the underlying security is 110?

A) 1.00.

B) 0.01.

C) 0.10.

Explanation

The gamma of an option is computed as follows:

Gamma = change in delta/change in the price of the underlying = (0.7 – 0.6)/(110 – 100) = 0.01

(Module 31.7, LOS 31.k)

Question #32 of 111

Question ID: 1473763

Which of the following *best* explains the sensitivity of a call option's value to volatility? Call option values:

A) increase as the volatility of the underlying asset increases because call options have limited risk but unlimited upside potential.

B) are not affected by changes in the volatility of the underlying asset.

C) increase as the volatility of the underlying asset increases because investors are risk seekers.

Explanation

A higher volatility makes it more likely that options end up in the money and can be exercised profitably, while the down side risk is strictly limited to the option premium.

(Module 31.7, LOS 31.n)

Question #33 of 111

Question ID: 1473764

Which of the following *best* describes the implied volatility method for estimated volatility inputs for the Black-Scholes model? Implied volatility is found:

A) using the most current stock price data.

B) using historical stock price data.

C) by solving the Black-Scholes model for the volatility using market values for the stock price, exercise price, interest rate, time until expiration, and option price.

Explanation

Implied volatility is found by "backing out" the volatility estimate using the current option price and all other values in the Black-Scholes model.

(Module 31.7, LOS 31.n)

Question #34 of 111

Question ID: 1473708

Pete Jenkins makes the following statement about options:

"$N(d_2)$ is interpreted as the risk-neutral probability that a call option will expire in the money. Similarly, $N(-d_2)$ is the risk-neutral probability that a put option will expire in the money."

Jenkins is *most likely*:

A) incorrect about the risk-neutral probability of put option expiring in the money.

B) correct.

C) incorrect about the risk-neutral probability of call option expiring in the money.

Explanation

Jenkins is correct about both probabilities.

(Module 31.6, LOS 31.g)

Question #35 of 111

Question ID: 1473762

If we use four of the inputs into the Black-Scholes-Merton option-pricing model and solve for the asset price volatility that will make the model price equal to the market price of the option, we have found the:

A) historical volatility.

B) option volatility.

C) implied volatility.

Explanation

The question describes the process for finding the expected volatility implied by the market price of the option.

(Module 31.7, LOS 31.n)

Question #36 of 111

Question ID: 1473691

Combining a short position in a stock with a long position in a call option on the stock will produce a payoff pattern equivalent to a:

A) long position in a put option on the stock.

B) short position in a put option on the stock.

C) risk-free security.

Explanation

The combined payoff pattern of a short position in a stock and a long call option on the stock is the same as the payoff pattern of a long put option on the stock.

(Module 31.2, LOS 31.c)

Question #37 of 111

Question ID: 1473686

Referring to put-call parity, which one of the following alternatives would allow you to create a synthetic riskless pure-discount bond?

A) Sell a European put option; sell the same stock; buy a European call option.

B) Buy a European put option; sell the same stock; sell a European call option.

C) Buy a European put option; buy the same stock; sell a European call option.

Explanation

According to put-call parity we can write a riskless pure-discount bond position as:

$$X/(1+R_f)^T = P_0 + S_0 - C_0$$

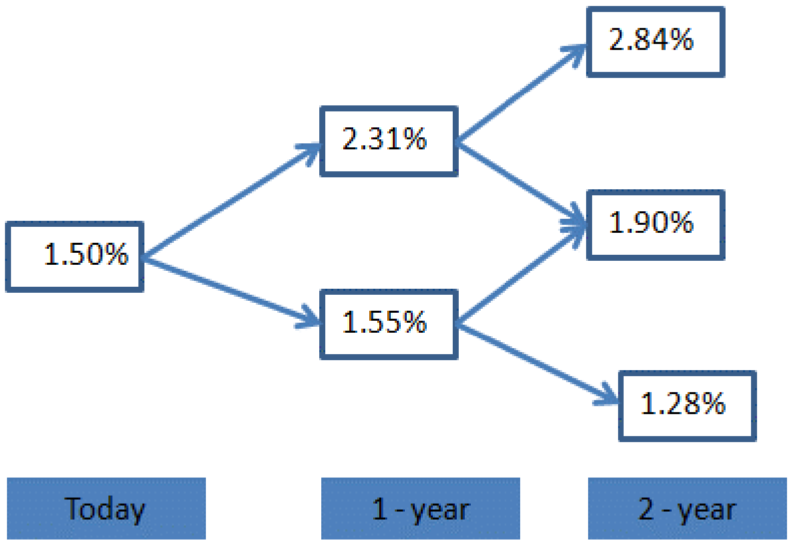
We can then read off the right-hand side of the equation to create a synthetic position in the riskless pure-discount bond. We would need to buy the European put, buy the same underlying stock, and sell the European call.

(Module 31.1, LOS 31.a)

Question #38 of 111

Question ID: 1473697

Given the following interest rate tree:



The value of a 2-period European put option with strike rate of 2% and notional principal of $1 million is *closest* to:

A) $3,109

B) $2,230

C) $2,020

Explanation

Given the exercise rate of 2.00%, the put option has a positive payoff for nodes P^{--} and P^{-+}.

The payoff at node P^{--} can be calculated as:

[Max (0, 0.02 – 0.0128)] × $ 1,000,000 = $7,200.

The payoff at node P^{-+} can be calculated as:

[Max (0, 0.02 – 0.019)] × $ 1,000,000 = $ 1,000.

Value at node P^{-} = [(0.5 × 7,200) + (0.5 × 1,000)] / (1.0155) = $4,037

Value at node P^{+} = [(0.5 × 0) + (0.5 × 1,000)] / (1.0231) = $489

And the value at node P = [(0.5 × 4,037) + (0.5 × 489)] / (1.015) = $2,230.

(Module 31.5, LOS 31.e)

Question #39 of 111

Question ID: 1586298

Dividends on a stock can be incorporated into the valuation model of an option on the stock by:

A) subtracting the present value of the dividend from the current stock price. ✓

B) adding the present value of the dividend to the current stock price. ✗

C) subtracting the future value of the dividend from the current stock price. ✗

Explanation

The option pricing formulas can be adjusted for dividends by subtracting the present value of the expected dividend(s) from the current asset price.

(Module 31.1, LOS 31.d)

Question #40 of 111

Question ID: 1473707

Bob Dilla, CFA makes the following statement about call options:

"Call options on stock can be thought of as leveraged stock investment where $N(d_1)$ units of stock is purchased using $e^{-rT}XN(-d_2)$ of borrowed funds."

Dilla is *most likely*:

A) correct.

B) incorrect about use of $e^{-rT}XN(-d_2)$ borrowed funds.

C) incorrect about $N(d_1)$ units of stock.

Explanation

Dilla is correct about calls being similar to a leveraged investment in $N(d_1)$ units of stock but is incorrect about the quantity of borrowed funds. It should be $e^{-rT}XN(d_2)$.

(Module 31.6, LOS 31.g)

Question #41 of 111

Question ID: 1473751

Which of the following statements concerning vega is *most* accurate? Vega is greatest when an option is:

A) far out of the money.

B) far in the money.

C) at the money.

Explanation

When the option is at the money, changes in volatility will have the greatest effect on the option value.

(Module 31.7, LOS 31.k)

Question #42 of 111

Question ID: 1542002

A stock is priced at 40 and the periodic risk-free rate of interest is 8%. The value of a two-period European call option with a strike price of 37 on a share of stock using a binomial model with an up factor of 1.20 and down factor of 0.833 is *closest* to:

A) $9.13.

B) $9.25.

C) $3.57.

Explanation

First, calculate the probability of an up move or a down move:

P_u = (1 + 0.08 − 0.833) / (1.20 − 0.833) = 0.673

P_d = 1 − 0.673 = 0.327

Two up moves produce a stock price of 40 × 1.44 = 57.60 and a call value at the end of two periods of 20.60. An up and a down move leave the stock price unchanged at 40 and produce a call value of 3. Two down moves result in the option being out of the money. The value of the call option is discounted back one year and then discounted back again to today. The calculations are as follows:

C^+ = [20.6(0.673) + 3(0.327)] / 1.08 = 13.745

C^- = [3(0.673) + 0 (0.327)] / 1.08 = 1.869

Call value today = [13.745(0.673) + 1.869(0.327)] / 1.08 = 9.13

(Module 31.1, LOS 31.b)

Question #43 of 111

Question ID: 1473693

Zetion Inc stock (current price $28) has 1-year call options with an exercise price of $30 trading at $2.07. The stock can increase by 15% or decrease by 13% over the next year and the risk-free rate is 3%. Arbitrage profits are *most likely*:

A) not possible. ✗

B) possible by purchasing 28 shares and writing 100 calls. ✓

C) possible by purchasing 100 calls and short selling 28 shares. ✗

Explanation

S^+ = 28(1.15) = 32.20; S^- = 28(0.87) = 24.36. C^+ = 32.20 – 30 = 2.20, C^- = 0.

H = (2.20 – 0) / (32.20 – 24.36) = 0.281

$$C_0 = hS_0 + \frac{(-hS^+ + C^+)}{(1+R_f)}$$

= 0.281 x 28 + [(- 0.281 x 32.20 + 2.20) / 1.03] = 1.22

Since the call price of $2.07 > 1.22, an arbitrage profit can be earned by writing calls and purchasing 0.281 shares per call written.

(Module 31.4, LOS 31.c)

Ronald Franklin, CFA, has recently been promoted to junior portfolio manager for a large equity portfolio at Davidson-Sherman (DS), a large multinational investment-banking firm. He is specifically responsible for the development of a new investment strategy that DS wants all equity portfolio managers to implement. Upper management at DS has instructed its portfolio managers to begin overlaying option strategies on all equity portfolios. The relatively poor performance of many of their equity portfolios has been the main factor behind this decision. Prior to this new mandate, DS portfolio managers had been allowed to use options at their own discretion, and the results were somewhat inconsistent. Some portfolio managers were not comfortable with the most basic concepts of option valuation and their expected return profiles, and simply did not utilize options at all. Upper management of DS wants Franklin to develop an option strategy that would be applicable to all DS portfolios regardless of their underlying investment composition. Management views this new implementation of option strategies as an opportunity to either add value or reduce the risk of the portfolio.

Franklin gained experience with basic options strategies at his previous job. As an exercise, he decides to review the fundamentals of option valuation using a simple example. Franklin recognizes that the behavior of an option's value is dependent on many variables and decides to spend some time closely analyzing this behavior. His analysis has resulted in the information shown in Exhibit 1 and Exhibit 2 for European style options.

Exhibit 1: Input for European Options

Stock Price (S)	100
Strike Price (X)	100
Interest Rate (r)	0.07
Dividend Yield (q)	0.00
Time to Maturity (years) (t)	1.00
Volatility (Std. Dev.)(Sigma)	0.20
Black-Scholes Put Option Value	$4.7809

Exhibit 2: European Option Sensitivities

Sensitivity	Call	Put
Delta	0.6736	–0.3264
Gamma	0.0180	0.0180
Theta	–3.9797	2.5470
Vega	36.0527	36.0527
Rho	55.8230	–37.4164

Question #44 - 47 of 111

Question ID: 1586289

Using the information in Exhibit 1, Franklin wants to compute the value of the corresponding European call option. Which of the following is the *closest* to Franklin's answer?

A) \$11.54.

B) \$4.78.

C) \$5.55.

Explanation

This result can be obtained using put-call parity in the following way:

$$\text{Call Value} = \text{Put Value} - Xe^{-rt} + S = \$4.78 - \$100.00e^{(-0.07 \times 1.0)} + 100 = \$11.54$$

The incorrect value of \$4.78 does not discount the strike price in the put-call parity formula.

(Module 31.1, LOS 31.a)

Question #45 - 47 of 111

Question ID: 1586290

Franklin wants to know how the put option in Exhibit 1 behaves when all the parameters are held constant except the delta. Which of the following is the *best* estimate of the change in the put option's price when the underlying equity increases by \$1?

A) −\$0.37.

B) −\$0.33.

C) −\$3.61.

Explanation

The correct value is simply the delta of the put option in Exhibit 2.

The incorrect value −\$3.61 represents the change due to the volatility divided by 10 multiplied by −1.

The incorrect value −\$0.37 calculates the change by dividing the short-term interest rate divided by 100.

(Module 31.1, LOS 31.a)

Question #46 - 47 of 111

Question ID: 1586291

Franklin computes the rate of change in the European put option delta value, given a $1 increase in the underlying equity. Using the information in Exhibit 1 and Exhibit 2, which of the following is the *closest* to Franklin's answer?

A) −0.3264.

B) 0.6736.

C) 0.0180.

Explanation

The correct value 0.0180 is referred to as the put option's Gamma.

The incorrect value −0.3264 is the delta of the put option.

The incorrect value 0.6736 is the call option's delta.

(Module 31.1, LOS 31.a)

Question #47 - 47 of 111

Question ID: 1586292

Franklin wants to know if the option sensitivities shown in Exhibit 2 have minimum or maximum bounds. Which of the following are the minimum and maximum bounds, respectively, for the put option delta?

A) −1 and 1.

B) There are no minimum or maximum bounds.

C) −1 and 0.

Explanation

The lower bound is achieved when the put option is far in the money so that it moves exactly in the opposite direction as the stock price. When the put option is far out of the money, the option delta is zero. Thus, the option price does not move even if the stock price moves since there is almost no chance that the option is going to be worth something at expiration.

(Module 31.1, LOS 31.a)

Question #48 of 111

Question ID: 1473732

Suppose a forward rate agreement (FRA) requires us to exchange six-month MRR one year from now for a fixed rate of interest of 8%. In other words, we will pay floating and receive fixed. Which of the following structures is *equivalent* to this FRA? A long:

A) call and a short put on MRR with a strike rate of 8% and six months to expiration.

B) call and a short put on MRR with a strike rate of 8% and twelve months to expiration.

C) put and a short call on MRR with a strike rate of 8% and twelve months to expiration.

Explanation

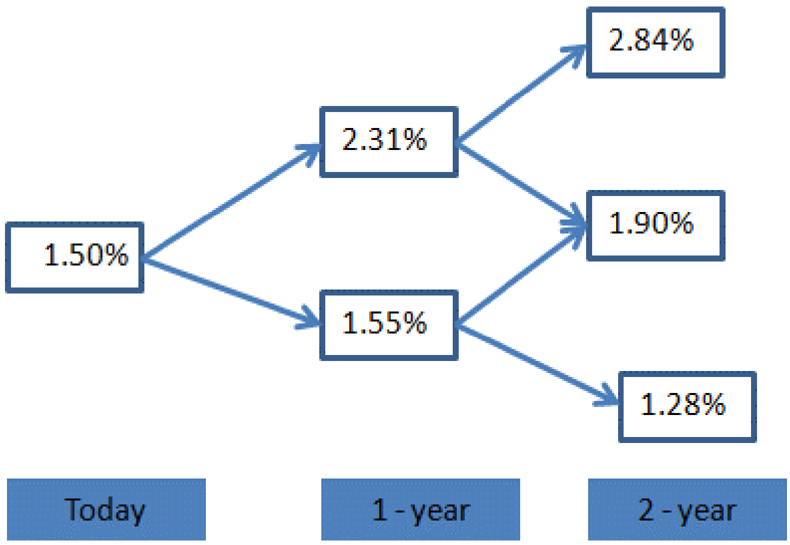
Interest rate swaps can be replicated with a series of put and call positions with expiration dates on the payment dates of the swap. For a receiver swap (where we pay floating and receive fixed), we need an option position that pays when floating rates fall and that requires a payment to be made when rates increase. A long interest rate put plus a short interest rate call would accomplish this. The strike rate of the options corresponds to the fixed rate of the FRA. The expiration of the option coincides with the MRR determination date.

(Module 31.6, LOS 31.j)

Question #49 of 111

Question ID: 1473696

Given the following interest rate tree:



The value of a 2-period European call option with strike rate of 2% and notional principal of $1 million is closest to:

A) $3,549

B) $2,022

C) $4,122

Explanation

Given the exercise rate of 2.00%, the call option has a positive payoff for node C^{++} only.

The payoff at node C^{++} can be calculated as:

[Max (0, 0.0284 – 0.02)] × $ 1,000,000 = $ 8,400.

Value at node C^{+} = [(0.5 × 8,400) + (0.5 × 0)] / (1.0231) = $4,105

Value at node C^{-} = 0

And the value at node C = [(0.5 × 4,105) + (0.5 × 0)] / (1.015) = $2,022

(Module 31.5, LOS 31.e)

Question #50 of 111

Question ID: 1586299

Early exercise of in-the-money American options on:

A) futures is sometimes worthwhile but never is for options on forwards.

B) forwards is sometimes worthwhile but never is for options on futures.

C) both futures and forwards is sometimes worthwhile.

Explanation

Early exercise of in-the-money American options on futures is sometimes worthwhile because the immediate mark to market upon exercise will generate funds that can earn interest. It is never worthwhile for options on forwards because no funds are generated until the settlement date of the forward contract.

(Module 31.1, LOS 31.d)

Question #51 of 111

Question ID: 1473739

A cap on a floating rate note, from the bondholder's perspective, is equivalent to:

A) owning a series of calls on fixed income securities.

B) writing a series of puts on fixed income securities.

C) writing a series of interest rate puts.

Explanation

For a bondholder, a cap, which puts a maximum on floating rate interest payments, is equivalent to writing a series of puts on fixed income securities. These would require the buyer to pay when rates rise and bond prices fall, negating interest rate increases above the cap rate. Writing a series of interest rate calls, not puts, would be an equivalent strategy. Calls on fixed income securities would pay when rates decrease, not when they increase.

(Module 31.6, LOS 31.j)

Jacob Bower is a bond strategist who would like to begin using fixed-income derivatives in his strategies. Bower has a firm understanding of the properties fixed-income securities. However, his understanding of interest rate derivatives is not nearly as strong. He decides to train himself on the valuation and sensitivity of interest rate derivatives using various interest rate scenarios. He considers the forward London Interbank Offered Rate (LIBOR) interest rate environment shown in Table 1. Using a rounded daycount (i.e., 0.25 years for each quarter) he has also computed the corresponding implied spot rates resulting from these LIBOR forward rates. These are included in Table 1.

Table 1: 90-Day LIBOR Forward Rates and Implied Spot Rates

Period (in months)	LIBOR Forward Rates	Implied Spot Rates

0 × 3	5.500%	5.5000%
3 × 6	5.750%	5.6250%
6 × 9	6.000%	5.7499%
9 × 12	6.250%	5.8749%
12 × 15	7.000%	6.0997%
15 × 18	7.000%	6.2496%

Bower has also estimated the LIBOR forward rate volatilities to be 20%. The particular fixed instruments that Bower would like to examine are shown in Table 2. He also wants to analyze the strategy shown in Table 3.

Table 2: Interest Rate Instruments

Dollar Amount of Floating Rate Bond			$42,000,000
Floating Rate Bond paying LIBOR +			0.25%
Time to Maturity (years)			8
Cap Strike Rate		7.00%	
Floor Strike Rate		6.00%	
Interest Payments		quarterly	

Table 3: Initial Position in 90-day LIBOR Eurodollar Contracts

Contract Month (from now)	Strategy A (contracts)	Strategy B (contracts)
3 months	300	100
6 months	0	100
9 months	0	100

Question #52 - 55 of 111

Question ID: 1586306

Bower is a bit puzzled about how to use caps and floors. He wonders how he could benefit both from increasing and decreasing interest rates. Which of the following trades would *most likely* profit from this interest rate scenario?

A) Buy at the money cap and sell at the money floor.

B) Sell at the money cap and at the money floor.

C) Buy at the money cap and at the money floor.

Explanation

This is a straddle on interest rates. The cap provides a positive payoff when interest rates rise and the floor provides a positive payoff when interest rates fall.

(Module 31.6, LOS 31.j)

Question #53 - 55 of 111

Question ID: 1586307

Bower has studied swaps extensively. However, he is not sure which of the following is the swap fixed rate for a one-year interest rate swap based on 90-day LIBOR with quarterly payments. Using the information in Table 1 and the formula below, what is the *most* appropriate swap fixed rate for this swap?

$$\mathrm{C} = \frac{1 - \mathrm{Z}_4}{\mathrm{Z}_1 + \mathrm{Z}_2 + \mathrm{Z}_3 + \mathrm{Z}_4}$$

where

$$\mathrm{Z}_n = \frac{1}{1 + \mathrm{R}_N} \text{price of } n - \text{zero} - \text{coupon bond per \$ of principal}$$

A) 5.75%.

B) 5.65%.

C) 6.01%.

Explanation

The swap fixed rate is computed as follows:

$$Z_{90-\text{day}} = \frac{1}{1+(0.055\times 90/360)}$$

$$= 0.98644$$

$$Z_{180-\text{day}} = \frac{1}{1+(0.05625\times 180/360)}$$

$$= 0.97264$$

$$Z_{270-\text{day}} = \frac{1}{1+(0.57499\times 270/360)}$$

$$= 0.95866$$

$$Z_{360-\text{day}} = \frac{1}{1+(0.058749\times 360/360)}$$

$$= 0.94451$$

$$\text{The quarterly fixed rate on the swap} = \frac{1-0.94451}{0.98644+0.97264+0.95866+0.94451}$$

$$= 0.05549/3.86225 = 0.01437 = 1.437\%$$

The fixed rate on the swap in annual terms is:

1.437% × 360 / 90 = 5.75%

(Module 31.6, LOS 31.f)

Question #54 - 55 of 111

Question ID: 1586308

Bower computes the implied volatility of a one year caplet on the 90-day LIBOR forward rates to be 18.5%. Using the given information what does this mean for the caplet's market price relative to its theoretical price? The caplet's market price is:

A) overvalued.

B) undervalued.

C) undervalued or overvalued.

Explanation

Volatility and option prices are always positively related. Therefore, since the option implied volatility is lower than the estimated volatility, this implies that the caplet is undervalued relative to its theoretical value.

(Module 31.6, LOS 31.f)

Question #55 - 55 of 111

Question ID: 1586309

For this question only, assume Bower expects the currently positively sloped LIBOR curve to shift upward in a parallel manner. Using a plain vanilla interest rate swap, which of the following will allow Bower to best take advantage of his expectations? Purchase a:

A) pay fixed interest rate swap. ✔

B) floating rate bond and enter into a receive fixed swap. ✘

C) receive fixed interest rate swap. ✘

Explanation

Since the interest rates are expected to rise for all maturities, one can benefit from this rise by receiving a floating rate (LIBOR) and borrowing at a fixed rate (i.e. a pay fixed swap).

(Module 31.6, LOS 31.f)

Question #56 of 111

Question ID: 1473733

The writer of a receiver swaption has:

A) an obligation to enter a swap in the future as the fixed-rate payer. ✔

B) an obligation to enter a swap in the future as the floating-rate payer. ✘

C) the right to enter a swap in the future as the floating-rate payer. ✘

Explanation

A receiver swaption gives its owner the right to receive fixed, the writer has an obligation to pay fixed.

(Module 31.6, LOS 31.j)

Nathan Detroit, a speculator, has come to you for technical advice regarding the pricing of swaps. He hopes to make big money in the swaps market from the exploitation of pricing discrepancies, but lacks an understanding of the principles underlying the pricing of swaps.

He asks you to consider a two-year, fixed-for-fixed, currency swap with semiannual payments. The domestic currency is the U.S. dollar and the foreign currency is the U.K. pound. The current exchange rate is $1.60 per pound. You forecast that the exchange rate would be $1.41 on the first settlement date. The notional principal of the swap is set at $10 million. The USD and £ term structure are shown in Exhibits 1 and 2 below.

Exhibit 1: Current USD Term Structure

Number of Days	MRR	Present Value Factor
180	0.0585	0.9716
360	0.0605	0.9430
540	0.0596	0.9179
720	0.0665	0.8826

Exhibit 2: Current £ Term Structure

Number of Days	MRR	Present Value Factor
180	0.0493	0.9759
360	0.0450	0.9569
540	0.0519	0.9278
720	0.0551	0.9007

Detroit has heard about the European put-call parity theorem and believes a synthetic call can be created through the use of a European put with the same strike as the call, a zero coupon bond with a face value equal to the strike price of the put and an underlying asset relating to the put and the call.

Detroit makes two comments regarding the BSM model and the Black model as follows:

Comment 1:	$N(d_1)$ in the BSM is the probability that the option will expire in the money.
Comment 2:	The probability that a receiver swaption will expire in the money is $N(-d_2)$.

Question #57 - 60 of 111

Question ID: 1473728

Calculate the USD swap fixed rate.

A) 3.16%.

B) 6.20%.

C) 6.32%.

Explanation

For a swap with n settlement periods, the semi-annual fixed payment per $1 of notional principal is calculated as:

$$\frac{1 - \text{PV factor for last cash flow}}{\sum\limits_{\text{i=1 to n}} \text{PV factor for ith cash flow}} = \frac{1-0.8826}{0.9716+0.9430+0.9180+0.8826} = 0.0316$$

The annualized fixed payment per $1 of notional principal is therefore:

$0.0316 \times \frac{360}{180} = 0.0632$ or 6.32%

(Module 30.6, LOS 30.e)

Question #58 - 60 of 111

Question ID: 1473729

Using the information in Exhibits 1 and 2, which of the following is *closest* to the amount of the British pound paid on the first settlement date?

A) £165,000.

B) £187,234.

C) £264,000.

Explanation

The semi-annual fixed payment per pound of notional principal is calculated as:

$$\frac{1-\text{PV factor for last cash flow}}{\sum\limits_{\text{i=1 to n}} \text{PV factor for ith cash flow}} = \frac{1-0.9007}{0.9759+0.9519+0.9278+0.9057} = 0.0264$$

Given the dollar notional principal of $10 million, the notional principal in British pounds (using the exchange rate on the date the swap is initiated) will be:

$$\frac{\$10\text{ million}}{\$1.60} = \$6.25\text{m}$$

Hence the payment made every 6 months will be £6.25m × 0.0264 = £165,000.

(Module 30.7, LOS 30.f)

Question #59 - 60 of 111

Question ID: 1473730

Which of the following would create a synthetic call?

A) Buy the put and the underlying. ✓

B) Sell the put, buy the underlying. ✗

C) Sell the put and the underlying. ✗

Explanation

When dealing with questions based on synthetics the solution can be arrived at by using put call parity.

Recall put + share = call + present value of bond PV(X)

The European put-call parity states that the payoffs of a fiduciary call (long zero coupon bond and long call) are equivalent to the payoffs of a protective put (long underlying and long put).

So a synthetic call = put + stock – PV(X)

We can then ignore the value of the pure discount bond PV(X) because its value is unaffected by the stock's price, leaving us with put + stock.

(Module 31.1, LOS 31.a)

Question #60 - 60 of 111

Question ID: 1473731

How many of Nathan's comments are correct?

A) Neither comment is correct.

B) Only one statement is correct.

C) Both statements are correct.

Explanation

$N(d_2)$ in both BSM and the Black model is the probability of the underlying asset being greater than the strike price at expiry. In the case of a swaption this would be the rate on a payer swap at expiry of the swaption being greater than the strike price (fixed rate) of the payer swaption.

$N(d_1)$ is a little more complicated than $N(d_2)$. It adjusts the current value of the underlying asset to represent the expected value of the underlying at expiry ignoring any values less that the strike price and factor in the probability of exercise. In practice this makes $N(d_1)$, a conditional probability, slightly bigger than $N(d_2)$ and unconditional probability. For the exam you should focus on knowing that $N(d_2)$ is the probability of exercise and that $N(d_1)$ is slightly higher. Do not waste too much time trying to understand exactly what $N(d_1)$ is doing.

If the $N(d_2)$ is the probability that ST > X at expiry (i.e., a call option is exercised) then 1 – $N(d_2)$ must be the probability that ST < X at expiry (i.e., the probability that a put option is exercised). It is common to write $N(-d_2)$ rather than 1 – (Nd_2).

(Module 31.6, LOS 31.j)

Lucy Wang is the Chief Financial Officer of Sam Corporation. Sam Corporation has floating rate liabilities and wants to hedge against the possibility of rising interest rates. Wang is looking into using swaptions to hedge against interest rate risk.

The board of Sam Corporation are not familiar with both swaps and swaptions. To explain the characteristics of swaps, Wang explains to the board that swaps are similar to interest rate options.

Wang decides to use a swaption to hedge against interest rate risk. She knows that there are two types of swaptions just like there are both call and put options, and that the cash flows on swaps can be replicated using swaptions.

Lucy is very interested in the application of the Black model in pricing swaptions. After a quick search online she has found the following:

$$\text{pay} = (AP)PVA\left[\left[SFR \times N(d_1) - X \times N(d_2)\right]\right] \times NP$$

where:

pay = value of the payer swaption

AP = 1 / number of settlement periods per year in the underlying swap

X = exercise rate specified in the swaption

NP = notional principal

Lucy is confused regarding what the notation PVA stands for and states:

"There are multiple payoffs on a swaption, each being the difference between the market swap rate at expiration and the exercise rate at each settlement date, over the swaps life. Given each payoff arises at a different point in time over the swaps life, each must be discounted back to the current period using discount rate specific to when it occurs. The PVA therefore must be an annuity factor summing all these specific discount rates."

Lucy also states:

"A payer swaption is right to enter a swap with a fixed rate equal to the strike price at the options maturity. The payoffs will therefore represent the difference in the swaptions strike price and the fixed market swap rate at the time of option expiry. Given we are comparing the payer swap underlying the swaption, versus the market rate of a payer swap at the option maturity, the floating payments can be ignored as they will offset. This is why the Black model only includes values for the current market swap fixed rate and fixed rate of the swap underlying the swaption (i.e., the strike)."

Question #61 - 64 of 111

Question ID: 1473718

Which of the following *best* describes how a payer swap could be replicated using a package of interest rate options?

A) The swap can be replicated by *buying* a package of interest rate *call* options *and* selling a package of interest rate put options at *different* strikes. ✗

B) The swap can be replicated by *selling* a package of interest rate *call* options and *buying* a package of interest rate *put* options at the *same* strikes. ✗

C) The swap can be replicated by *buying* a package of interest rate *call* options and *selling* a package of interest rate *put* options at the same strikes. ✓

Explanation

The pay-fixed party in an interest rate swap receives a net payment if the floating rate is above the fixed rate and pays when the floating rate is below the fixed rate. This payoff characteristic is similar to buying a package of interest rate call options (receive payment when reference rate is above the strike rate) and selling a package of interest rate put options (pays when the reference rate is below the strike rate) at the same strikes.

(Module 31.6, LOS 31.j)

Question #62 - 64 of 111

Question ID: 1473719

Which of the following *best* describes how a payer swap could be replicated using interest rate swaptions?

A) The swap can be replicated by selling a payer swaption and buying a receiver swaption at the same strike. ✗

B) The swap can be replicated by buying a payer swaption and selling a receiver swaption at the same strike. ✓

C) The swap can be replicated by buying a payer swaption and selling a receiver swaption at different strikes. ✗

Explanation

The swap can be replicated by buying a payer swaption and selling a receiver swaption at the same strikes.

Payer swaption is the right to enter the swap as the fixed-rate payer and receiver swaption is the right to enter as fixed-rate receiver. Payer swaption increases in value when the floating rate increases, similar to that for the pay-fixed party in an interest rate swap. Receiver swaption increases in value when the floating rate decreases. To replicate the negative value suffered by the pay-fixed interest rate party when floating rate decreases, the swap can be replicated by selling a receiver swaption as the negative value increases when rates fall.

(Module 31.6, LOS 31.j)

Question #63 - 64 of 111

Question ID: 1473720

With regards to Lucy's comments regarding the PVA in the Black model and the offsetting of floating rates at expiry of the swaption, which of the following is *most* accurate?

A) Lucy is correct about the definition of the PVA and the offsetting of floating rates.

B) Lucy is correct about the offsetting of the floating rates but not the PVA.

C) Lucy is correct about the PVA but not the offsetting of floating rates.

Explanation

Wang is correct about both parts.

Exercise of a payer swaption results in a swap with fixed payments equal to the strike price on the swaption and the receipt of floating payments.

The payoff on a swaption is the difference between the fixed swap rate that can be achieved by exercising the option and the market fixed rate on a swap at expiry. This payoff will be received at every settlement date over the swaps life. The PVA is therefore an annuity factor, which will discount all the payoffs back to the current period.

The market swap at expiry of the swaption is pay fixed at the market rate and receive floating payments. The swaption can be exercised into a pay fixed (at swaption strike) and receive floating swap. The floating payments are identical on the two swaps and therefore can be ignored for valuation purposes.

(Module 31.6, LOS 31.j)

Question #64 - 64 of 111

Question ID: 1473721

Which of the following comments relating to the Black model valuation of a swaption is the *most* accurate?

A) SFR in the formula identified by Lucy is the market swap fixed rate at the expiration of the swaption.

B) $N(d_2)$ is likely to be greater than $N(d_1)$.

C) A payer swaption will only be exercised in the market swap fixed rate at expiry is greater than the swaption's strike price.

Explanation

SFR in the Black formula represents the current market swap fixed rate not the fixed rate at expiration. Both the Black and BSM formulas use the current price of the underlying asset.

The Black formula, like BSM computes the expected value of what you will receive less the present value of what you will pay given assumptions such as volatility.

$N(d_2)$ = only considers probability that the option is exercised (i.e., the probability of the market SFR being greater than the strike at expiration). $N(d_1)$ has to adjust the current SFR to the expected SFR at expiration given that the option will only be exercised if the market SFR > strike rate. The result is that in the Black and BSM formulas that $N(d_1)$ is slightly higher than $N(d_2)$.

A payer swaption will only be exercised if the strike rate is lower than the market rate on a pay fixed swap at expiry. When a swaption is exercised the holder will pay fixed at the strike price and receive floating. This will be attractive when the fixed rate on a pay fixed swap in the market is higher. The floating payments can be ignored as the rate on swaption when exercised and the rate on a new swap at expiry would be identical.

(Module 31.6, LOS 31.j)

Question #65 of 111

Question ID: 1473746

For a change in which of the following inputs into the Black-Scholes-Merton option pricing model will the direction of the change in a put's value and the direction of the change in a call's value be the same?

A) Volatility. ✓

B) Exercise price. ✗

C) Risk-free rate. ✗

Explanation

A decrease/increase in the volatility of the price of the underlying asset will decrease/increase both put values and call values. A change in the values of the other inputs will have opposite effects on the values of puts and calls.

(Module 31.7, LOS 31.k)

Question #66 of 111

Question ID: 1473709

Compared to the value of a call option on a stock with no dividends, a call option on an identical stock expected to pay a dividend during the term of the option will have a:

A) higher value only if it is an American style option.

B) lower value in all cases.

C) lower value only if it is an American style option.

Explanation

An expected dividend during the term of an option will decrease the value of a call option.

(Module 31.6, LOS 31.h)

Question #67 of 111

Question ID: 1473745

The value of a European call option on an asset with no cash flows is positively related to all of the following EXCEPT:

A) exercise price.

B) risk-free rate.

C) time to exercise.

Explanation

The value of a call option decreases as the exercise price increases.

(Module 31.7, LOS 31.k)

Question #68 of 111

Question ID: 1473690

A non-dividend-paying option on a stock is *most likely* to be exercised early if the option is a(n):

A) European option.

B) call option.

C) put option.

Explanation

After exercising a deep in-the-money put early, the sale proceeds can be invested at the risk-free rate, and in this way the investor may earn interest worth more than the time value of the put. Non-dividend-paying call options on stock will never be exercised early because the minimum price of the option always exceeds its exercise value. European options cannot be exercised early.

(Module 31.1, LOS 31.d)

Question #69 of 111

Question ID: 1473737

A floor on a floating rate note, from the bondholder's perspective, is equivalent to:

A) owning a series of puts on fixed income securities.

B) writing a series of interest rate puts.

C) owning a series of calls on fixed income securities.

Explanation

A floor, which puts a minimum on floating rate interest payments is equivalent to owning calls on fixed income securities which will pay when interest rates fall. Owning interest rate puts, rather than writing them, would be equivalent to the floor. Puts on fixed income securities pay when interest rates increase.

(Module 31.6, LOS 31.j)

Question #70 of 111

Question ID: 1473765

In order to compute the implied asset price volatility for a particular option, an investor:

A) must have the market price of the option.

B) must have a series of asset prices.

C) does not need to know the risk-free rate.

Explanation

In order to compute the implied volatility we need the risk-free rate, the current asset price, the time to expiration, the exercise price, and the market price of the option.

(Module 31.7, LOS 31.n)

Susan Smith is analyzing the stock of FDL Inc. She has found the following quotations (all prices in dollars per share) on 91-day European options:

Exhibit 1

Option Strike Price	Call Premium	Put Premium
50	5.28	1.54

55	2.64	3.33
60	1.14	X
Other Information		
Risk-free interest rates	6%	
Annual volatility	30%	
FDL Inc share price	$53	

*FDL Inc currently does not pay dividends

Smith wants to value the equity call options of FDL Inc. using the Black-Scholes-Merton (BSM) model. She wants to understand the assumptions and the limitations of the model and asks David Wang for help. Wang provides the information shown in Exhibits 2 and 3.

Exhibit 2

Assumptions of BSM Model
Assumption 1
The underlying asset returns follow a normal distribution
Assumption 2
Options are European style

Exhibit 3

Assumptions of BSM Model	Implications
The risk-free rate is known and constant over the options life	**Implication 1** Useful for pricing options on bond prices and interest rates

The continuously compounded yield on the asset is constant	**Implication 2** BSM model can be modified to account for cash flows on the underlying

Question #71 - 74 of 111

Question ID: 1473699

Using information in Exhibit 1, the value of $60 strike put option is *closest* to:

A) $4.99.

B) $5.86.

C) $7.27.

Explanation

The put-call parity equation is: $C + X / (1 + r)^t = P + S$

Hence $P = C – S + X / (1 + r)^t = 1.14 – 53 + 60 / (1.06)^{91/365} = \$7.27$

(Module 31.1, LOS 31.a)

Question #72 - 74 of 111

Question ID: 1473700

Susan discovers that the fair value for the $55 strike put is in fact $3.85. Which of the following is the *most* appropriate set of transactions to exploit the mispricing (ignore transaction costs)?

A) Write a call option, buy a put option, buy one share, borrow the PV of strike.

B) Buy a call option, write a put option, buy one share, borrow the PV of strike.

C) Buy a call option, write a put option, sell short one share, put on deposit the PV of strike.

Explanation

The \$55-strike put is currently underpriced compared to its synthetic (\$3.33 < \$3.85).

We therefore will want to go long the underpriced put and short the overpriced synthetic to make an arbitrage profit.

Susan will be obliged to sell the shares at \$55, so she also needs to buy the stock (so that she has it to deliver) and borrow the PV of the \$55 strike (\$55 / $(1.06)^{91/365}$ = \$54.21). The net result will be a receipt of \$2.64 – \$3.33 – \$53 + 54.21 = \$0.52 per share. Thus:

Write a call option, buy a put option, buy one share, and borrow the PV of strike.

(Module 31.1, LOS 31.a)

Question #73 - 74 of 111

Question ID: 1477072

Using information in Exhibit 2, which of the following statements about assumptions 1 and 2 is *most* accurate?

A) Both assumptions are correct.

B) Only one of the two assumptions is correct.

C) Both assumptions are incorrect.

Explanation

The assumptions used by the basic BSM model include:

- The underlying asset price follows a geometric Brownian motion process. The continuously compounded return is normally distributed. Under this framework, change in asset price is continuous: there are no abrupt jumps.
- The (continuously compounded) risk-free rate is constant and known. Borrowing and lending are both at the risk-free rate.
- The volatility of the returns on the underlying asset is constant and known.
- Markets are "frictionless." There are no taxes, no transactions costs, and no restrictions on short sales or the use of short-sale proceeds. Continuous trading is possible, and there are no arbitrage opportunities in the marketplace.
- The (continuously compounded) yield on the underlying asset is constant.
- The options are European options (i.e., they can only be exercised at expiration).

Assumption 1 is incorrect. Asset prices are lognormally distributed and asset returns are normally distributed.

Assumption 2 is correct.

(Module 31.6, LOS 31.f)

Question #74 - 74 of 111

Question ID: 1542004

Using Exhibit 3, which of the following statements about implications 1 and 2 is *most* accurate?

A) Both implications are correct.

B) Both implications are incorrect.

C) Only one of the two implications is correct.

Explanation

Implication 1 is incorrect. The BSM model assumes a constant risk-free interest rate but interest rate volatility is a key factor that determines the value of options on bonds and interest rate related contracts. Hence BSM is not useful for pricing options on bond prices and interest based derivatives.

Implication 2 is correct. BSM allows for constant continuously compounded dividend yield (i.e., cash flow on the underlying) by adjusting the asset value by the present value of the expected cash flows.

(Module 31.6, LOS 31.f)

Rachel Barlow is a recent graduate of Columbia University with a Bachelor's degree in finance. She has accepted a position at a large investment bank, but first must complete an intensive training program to gain experience in several of the investment bank's areas of operations. Currently, she is spending three months at her firm's Derivatives Trading desk. One of the traders, Jason Coleman, CFA, is acting as her mentor, and will be giving her various assignments over the three month period.

One of the first projects Coleman asks Barlow to do is to compare different option trading strategies. Coleman would like Barlow to pay particular attention to strategy costs and their potential payoffs. Barlow is not very comfortable with option models, and knows she needs to be able to fully understand the most basic concepts in order to move on. She decides that she must first investigate how to properly price European and American style equity options. Coleman has given Barlow software that provides a variety of analytical information using three valuation approaches: the Black-Scholes model, the Binomial model, and Monte Carlo simulation. Barlow has decided to begin her analysis using a variety of different scenarios to evaluate option behavior. The data she will be using in her scenarios is provided in Exhibits 1 and 2. Note that all of the rates and yields are on a continuous compounding basis.

Exhibit 1

Stock Price (S)	$100.00

Strike Price (X)	$100.00
Interest Rate (r)	7.0%
Dividend Yield (q)	0.0%
Time to Maturity (years)	0.5
Volatility (Std. Dev.)	20.0%
Value of Put	$3.9890

Exhibit 2

Stock Price (S)	$110.00
Strike Price (X)	$100.00
Interest Rate (r)	7.0%
Dividend Yield (q)	0.0%
Time to Maturity (years)	0.5
Volatility (Std. Dev.)	20.0%
Value of Call	$14.8445
$N(d_1)$	0.8394
$N(d_2)$	0.8025

Exhibit 3

Stock Price (S)	$115.00
Strike Price (X)	$100.00
Interest Rate (r)	7.0%
Dividend Yield (q)	0.0%
Time to Maturity (years)	0.5
Volatility (Std. Dev.)	20.0%
Value of Call	$19.2147
Value of Put	$0.7753

Question #75 - 78 of 111

Question ID: 1586294

Barlow notices that the stock in Exhibit 1 does not pay dividends. If the stock begins to pay a dividend, how will the price of a call option on that stock be affected? The price of the call option:

A) may either increase or decrease.

B) will decrease.

C) will increase.

Explanation

The call option value will decrease since the payment of dividends reduces the value of the underlying, and the value of a call is positively related to the value of the underlying.

(Module 31.1, LOS 31.a)

Question #76 - 78 of 111

Question ID: 1586295

Barlow calculated the value of an American call option on the stock shown in Exhibit 2. Which of the following is *closest* to the value of this call option?

A) $15.41.

B) $14.84.

C) $15.12.

Explanation

The value of the American-style call option is the same as the value of the equivalent European-style call option. Since the underlying stock does not pay a dividend, it is never optimal to exercise the American option early. Hence the early-exercise option embedded in the American-style call has no value in this case. This makes the American option worth exactly the same as the European option.

(Module 31.1, LOS 31.a)

Question #77 - 78 of 111

Question ID: 1586296

Using the information in Exhibit 2, Barlow computes the value of a European put option. Which of the following is *closest* to the value of this option?

A) $1.97.

B) $4.84.

C) $1.41.

Explanation

Using the information in Exhibit 2, this value can be determined from put-call parity as follows:

$$\text{Put} = \text{Call} + Xe^{-rt} - S$$

So we have Put = $14.8445 + $100.00e$^{(-7.00\% \times 0.5)}$ − $110.00 = $1.4050

(Module 31.1, LOS 31.a)

Question #78 - 78 of 111

Question ID: 1586297

Barlow notices that the stock in Exhibit 2 does not pay dividends. If the stock starts to pay a dividend, how will the price of a put option on that stock be affected?

A) Increase.

B) Decrease.

C) Increase or decrease.

Explanation

The put option value will increase since the payment of dividends reduces the value of the underlying, and the value of a put is negatively related to the value of the underlying.

(Module 31.1, LOS 31.a)

Question #79 of 111

Question ID: 1473692

To create a synthetic short position in a stock, an investor can buy:

A) a call option on the stock and sell a put option on the stock.

B) a put option on the stock and sell a call option on the stock.

C) both a call option on the stock and a put option on the stock.

Explanation

Buying a put option and writing a call option results in a payoff pattern similar to that of a short position in the underlying stock.

(Module 31.2, LOS 31.c)

Question #80 of 111

Question ID: 1473704

Which of the following is *least likely* one of the assumptions of the Black-Scholes-Merton option pricing model?

A) Changes in volatility are known and predictable.

B) The risk-free rate of interest is known and does not change over the term of the option.

C) The options are European.

Explanation

The BSM model assumes that volatility of the return is known and constant (i.e., not changing). Other assumptions of the BSM model include the continuously compounded risk-free interest rate is known and constant and the options are European (meaning that early exercise is not allowed).

(Module 31.6, LOS 31.f)

Question #81 of 111

Question ID: 1473694

DTK Inc stock (current price $55) has 1-year call options with an exercise price of $55 trading at $4.92. The stock can increase by 20% or decrease by 15% over the next year and the risk-free rate is 5%. Arbitrage profits are *most likely*:

A) not possible.

B) possible by purchasing 100 calls and short selling 57 shares.

C) possible by purchasing 57 shares and writing 100 calls.

Explanation

S^+ = 55(1.20) = 66; S- = 55 (0.85) = 46.75. C^+ = 66 – 55 = 11, C^- = 0.

H = (11 – 0) / (66 – 46.75) = 0.5714

$$C_0 = hS_0 + \frac{(-hS^+ + C^+)}{(1+R_f)}$$

= (0.571×55) + [((-0.571×66) + 11) / 1.05] = 5.99

Since the call price of $4.92 < $5.99, an arbitrage profit can be earned by buying calls and short selling 0.571 shares per call bought.

(Module 31.4, LOS 31.c)

Question #82 of 111

Question ID: 1473741

Which of the following *best* represents an interest floor?

A) A portfolio of put options on an interest rate.

B) A put option on an interest rate.

C) A portfolio of call options on an interest rate.

Explanation

A long floor (floor buyer) has the same general expiration-date payoff diagram as that for long interest rate put position.

(Module 31.6, LOS 31.j)

Joel Franklin, CFA, has recently been promoted to junior portfolio manager for a large equity portfolio at Davidson Sherman (DS), a large multinational investment banking firm. The portfolio is subdivided into several smaller portfolios. In general, the portfolios are composed of U.S. based equities, ranging from medium to large-cap stocks. Currently, DS is not involved in any foreign markets. In his new position, he will now be responsible for the development of a new investment strategy that DS wants all of its equity portfolios to implement. The strategy involves overlaying option strategies on its equity portfolios. Recent performance of many of their equity portfolios has been poor relative to their peer group. The upper management at DS views the new option strategies as an opportunity to either add value or reduce risk.

Franklin recognizes that the behavior of an option's value is dependent upon many variables and decides to spend some time closely analyzing this behavior. He took an options strategies class in graduate school a few years ago, and feels that he is fairly knowledgeable about the valuation of options using the Black-Scholes model. Franklin understands that the volatility of the underlying asset returns is one of the most important contributors to option value. Therefore, he would like to know when the volatility has the largest effect on option value. Upper management at DS has also requested that he further explore the concept of a delta neutral portfolio. He must determine how to create a delta neutral portfolio, and how it would be expected to perform under a variety of scenarios. Franklin is also examining the change in the call option's delta as the underlying equity value changes. He also wants to determine the minimum and maximum bounds on the call option delta. Franklin has been authorized to purchase calls or puts on the equities in the portfolio. He may not, however, establish any uncovered or "naked" option positions. His analysis has resulted in the information shown in Exhibit 1 and Exhibit 2 for European style options.

Exhibit 1: Input for European Options

Stock Price (S)	100
Strike Price (X)	100
Interest Rate (r)	0.07
Dividend Yield (q)	0
Time to Maturity (years) (t)	1
Volatility (Std. Dev.) (sigma)	0.2
Black-Scholes Put Option Value	$4.7809

Exhibit 2: European Option Sensitivities

Sensitivity	Call	Put
Delta	0.6736	−0.3264
Gamma	0.0180	0.0180
Theta	−3.9797	2.5470
Vega	36.0527	36.0527
Rho	55.8230	−37.4164

Question #83 - 86 of 111

Question ID: 1586321

Which of the following *most* accurately describes when the call option delta reaches its minimum bound? The call option reaches its minimum bound when call option is:

A) at the money.

B) the option's delta has no minimum bound.

C) far out of the money.

Explanation

When a call option is far out of the money its value is insensitive to changes in value of the underlying. This is because the chances that it is going to end up in the money at expiration are very small.

(Module 31.7, LOS 31.l)

Question #84 - 86 of 111

Question ID: 1586322

If the portfolio has 10,000 shares of the underlying stock and he wants to completely hedge the price risk using options, what kind of options should Franklin buy?

A) Call and put options.

B) Call options.

C) Put options.

Explanation

Buying put options will allow Franklin to completely hedge the stock price risk.

(Module 31.7, LOS 31.l)

Question #85 - 86 of 111

Question ID: 1586323

Compute the number of shares of stock necessary to create a delta neutral portfolio consisting of 100 long put options in Exhibit 2 and the stock.

A) 67.36.

B) 32.64.

C) −32.64.

Explanation

This is simply −100 times the put option delta. Since each share has a delta of 1, we only need 32.64 shares (long) to create a delta neutral portfolio.

(Module 31.7, LOS 31.l)

Question #86 - 86 of 111

Question ID: 1586324

Compute the number of shares of stock necessary to create a delta neutral portfolio consisting of 100 long call options in Exhibit 2 and the stock.

A) 67.36.

B) −67.36.

C) −32.64.

Explanation

This is simply −100 times the call option delta. Since each share has a delta of 1, we only need −67.36 (short) shares to create a delta neutral portfolio.

(Module 31.7, LOS 31.l)

Question #87 of 111

Question ID: 1473742

Suppose a forward rate agreement (FRA) calls for us to receive the six-month MRR two years from now for a payment of a fixed rate of interest of 6%. Which of the following structures is equivalent to this long FRA? A long:

A) put and a short call on MRR with a strike rate of 6% and two years to expiration.

B) call and a short put on MRR with a strike rate of 6% and two years to expiration.

C) call on MRR with a strike rate of 6% and eighteen months to expiration.

Explanation

A long FRA is replicated by a long IR call and short IR put with expiration corresponding to the FRA settlement date.

(Module 31.6, LOS 31.j)

Al Bingly, CFA, is a derivatives specialist who attempts to identify and make short-term gains from trading mispriced options. One of the strategies that Bingly uses is to look for arbitrage opportunities in the market for European options. This strategy involves creating a synthetic call from other instruments at a cost less than the market value of the call itself, and then selling the call. During the course of his research, he observes that Hilland Corporation's stock is currently priced at $56, while a European-style put option with a strike price of $55 is trading at $0.40 and a European-style call option with the same strike price is trading at $2.50. Both options have 6 months remaining until expiration. The risk-free rate is currently 4 percent.

Bingly often uses the binomial model to estimate the fair price of an option. He then compares his estimated price to the market price. He observes that Dale Corporation's stock has a current market price of $200, and he predicts that its price will either be $166.67 or $240 in one year. The risk-free rate is currently 4 percent. He also observes that the price of a one-year call with a $220 strike price is $11.11.

Bingly also uses the Black-Scholes-Merton model to price options. His stated rationale for using this model is that he believes the prices of the stocks he analyzes follow a lognormal distribution, and because the model allows for a varying risk-free rate over the life of the option. His plan is to use a statistical technique to estimate the volatility of a stock, enter it into the Black-Scholes-Merton model, and see if the associated price is higher or lower than the observed market price of the options on the stock.

Bingly wishes to apply the Black-Scholes-Merton model to both non-dividend paying and dividend paying stocks. He investigates how the presence of dividends will affect the estimated call and put price.

Question #88 - 91 of 111

Question ID: 1586301

The one-year call option on Dale Corporation:

A) is underpriced.

B) is overpriced.

C) may be over or underpriced. The given information is not sufficient to give an answer.

Explanation

The up movement parameter U=1.20, and the down movement parameter D=0.833. We calculate the probability of an up move π_U = (1 + 0.04 – 0.833)/(1.2 – 0.833) = 0.564. The call is out of the money in the event of a down movement, and has an intrinsic value of $20 in the event of an up movement. Therefore, the estimated value of the call is C = (0.564) × $20 / (1.04) = $10.85. Thus, the price of $11.11 is too high and the call is overpriced.

(Module 31.6, LOS 31.f)

Question #89 - 91 of 111

Question ID: 1586302

Bingly's sentiments towards the Black-Scholes-Merton (BSM) model regarding a lognormal distribution of prices and a variable risk-free rate are:

A) correct for both reasons.

B) correct concerning the distribution of stocks but incorrect concerning the risk-free rate.

C) incorrect for both reasons.

Explanation

The model requires many assumptions, e.g., the distribution of stock prices is lognormal and the risk-free rate is known and *constant*. Other assumptions are frictionless markets, the options are European, and the volatility is known and constant.

(Module 31.6, LOS 31.f)

Question #90 - 91 of 111

Question ID: 1586303

If Bingly forecasts the volatility for a stock and find that it is significantly greater than that implied by the prices of the puts and calls of the stock, he would conclude that:

A) the puts are overpriced and the calls are underpriced. ✗

B) puts and calls are underpriced. ✓

C) puts and calls are overpriced. ✗

Explanation

There is a positive relationship between the volatility of the stock and the price of both puts and calls. A higher estimate of volatility implies that the prices of both puts and calls should be higher.

(Module 31.6, LOS 31.f)

Question #91 - 91 of 111

Question ID: 1586304

All else being equal, the greater the dividend paid by a stock the:

A) lower the call price and the higher the put price. ✓

B) lower the call price and the lower the put price. ✗

C) higher the call price and the lower the put price. ✗

Explanation

When dividend payments occur during the life of the option, the price of the underlying stock is reduced (on the ex-dividend date). All else being equal, the lower price reduces the value of call options and increases the value of put options.

(Module 31.6, LOS 31.f)

John Fairfax is a recently retired executive from Reston Industries. Over the years he has accumulated $10 million worth of Reston stock and another $2 million in a cash savings account. He hires Richard Potter, CFA, a financial adviser from Stan Morgan, LLC, to help him develop investment strategies. Potter suggests a number of interesting investment strategies for Fairfax's portfolio. Many of the strategies include the use of various equity derivatives. Potter explains to Fairfax that there are numerous options available for him to obtain almost any risk return profile he might need. Potter suggests that Fairfax consider options on both Reston stock and the S&P 500. Potter collects the information needed to evaluate options for each security. These results are presented in Table 1.

Table 1: Option Characteristics

	Reston	S&P 500
Stock price	$50.00	$1,400.00
Strike price	$50.00	$1,400.00
Interest rate	6.00%	6.00%
Dividend yield	0.00%	0.00%
Time to expiration (years)	0.5	0.5
Volatility	40.00%	17.00%
Beta Coefficient	1.23	1
Correlation	0.4	

Potter presents Fairfax with the prices of various options as shown in Table 2. Table 2 details standard European calls and put options. Potter presents the option sensitivities in Potter presents Fairfax with the prices of various options as shown in Table 3and Potter presents Fairfax with the prices of various options as shown in Table 4.

Table 2: Regular and Options (Option Values)

	Reston	S&P 500
European call	$6.31	$6.31
European put	$4.83	$4.83
American call	$6.28	$6.28
American put	$4.96	$4.96

Table 3: Reston Stock Option Sensitivities

	Delta
European call	0.5977

European put	−0.4023
American call	0.5973
American put	−0.4258

Table 4: S&P 500 Option Sensitivities

	Delta
European call	0.622
European put	−0.378
American call	0.621
American put	−0.441

Question #92 - 95 of 111

Question ID: 1586331

Given the information regarding the various Reston stock options, which option will increase the *most* relative to an increase in the underlying Reston stock price?

A) European call. ✓

B) American put. ✗

C) American call. ✗

Explanation

Using its delta in Potter presents Fairfax with the prices of various options as shown in Table 3, if the Reston stock increases by a dollar the European call on the stock will increase by 0.5977.

(Module 31.7, LOS 31.l)

Question #93 - 95 of 111

Question ID: 1586332

Fairfax would like to consider neutralizing his Reston equity position from changes in the stock price of Reston. Using the information in Table 3 how many standard Reston European options would have to be either bought or sold in order to create a delta neutral portfolio?

A) Sell 334,616 put options. ✗

B) Buy 300,703 put options. ✗

C) Sell 334,616 call options.

Explanation

Number of call options = (Reston Portfolio Value / Stock PriceReston)(1 / Deltacall).

Number of call options = ($10,000,000 / $50.00/sh)(1 / 0.5977) = 334,616.

(Module 31.7, LOS 31.l)

Question #94 - 95 of 111

Question ID: 1586333

Fairfax remembers Potter explaining something about how options are not like futures and swaps because their risk-return profiles are non-linear. Which of the following option sensitivity measures does Fairfax need to consider to completely hedge his equity position in Reston from changes in the price of Reston stock?

A) Gamma and Theta.

B) Delta and Gamma.

C) Delta and Vega.

Explanation

Vega measures the sensitivity relative to changes in volatility. Theta measures sensitivity relative to changes in time to expiration.

(Module 31.7, LOS 31.l)

Question #95 - 95 of 111

Question ID: 1586334

Fairfax has heard people talking about "making a portfolio delta neutral." What does it mean to make a portfolio delta neutral? The portfolio:

A) is insensitive to stock price changes.

B) is insensitive to interest rate changes.

C) is insensitive to volatility changes in the returns on the underlying equity.

Explanation

The delta of the option portfolio is the change in value of the portfolio if the stock price changes. A delta neutral option portfolio has a delta of zero.

(Module 31.7, LOS 31.l)

Gina Davalos, CFA is a portfolio manager for the Herron Investments. She is interested in hedging the equity risk of one of her clients, Lou Gier. Gier has 200,000 shares of a stock with the symbol QJX that he believes could take a dive in the next 9 months. Davalos gathers the following information to suggest potential strategies to offset the potential loss. Each option contract is for 100 options.

General Information:

QJX Current Stock Price	$100.00
Risk-free rate	5.0%
QJX Dividend Yield	0.0%
Time to Maturity (years)	0.75

Option Information:

Strike Price	$100.00
Value of Call	$12.09
Delta on Call Option	0.6081
Value of Put (years)	$8.41

Equity Swap Information:

Terms	9 months
Settlement frequency	Quarterly
Fixed rate	6.0%
Return on QJX	Variable

Futures Information:

Terms	9 months
Current Futures Price	$105.50

Question #96 - 99 of 111

Question ID: 1586326

In order to create a delta-neutral hedge using put option contracts, Davalos would *most accurately* need to:

A) buy 2,000 contracts.

B) buy 5,103 contracts.

C) sell 510,271 contracts.

Explanation

The delta of a put option is the delta of the corresponding call option minus- 1. The delta of a QJX put option is thus –0.3919. The number of put options needed is 200,000 / –0.3919 = –510,334 options or approximately 5,103 contracts per 100 shares. Gier is long the stock, to hedge with puts Davalos should also take a long position in the puts.

(Module 31.7, LOS 31.l)

Question #97 - 99 of 111

Question ID: 1586327

An equity swap to hedge the equity risk for Gier would result in a net return on the portfolio of a:

A) fixed rate of 4.5% for the year.

B) variable rate based on the total return of QJX stock.

C) fixed rate of 1.5% per quarter.

Explanation

To offset the equity risk, Gier would pay a variable rate based on the total return of QJX and receive a fixed rate. The quoted rate is an annualized rate and since the swap is for three quarters or nine months, the full 6.0% will not be realized. The 6.0% annualized rate is equivalent to 1.5% per quarter. Any return on the portfolio would be passed on to the swap counterparty and Gier will be immune to equity market movements.

(Module 31.7, LOS 31.l)

Question #98 - 99 of 111

Question ID: 1586328

If the equity swap is implemented and after 3 months the stock price has increased to $106.00, the net cash flow for the swap is:

A) a loss of $900,000.

B) a gain of $900,000.

C) zero.

Explanation

The equity swap requires Gier to pay a variable rate of total return on QJX and receive a fixed rate. If the stock appreciates, the swap results in a positive cash flow of 6.0%/4 × $20,000,000 = $300,000 and a negative cash flow of $20,000,000 × ($106/$100 – 1) = $1,200,000, summing to a net of outflow of $900,000. The swap plus the equity position result in an overall gain, as the gain on the stock more than offsets the loss on the equity swap.

(Module 31.7, LOS 31.l)

Question #99 - 99 of 111

Question ID: 1586329

Based on the futures information, an arbitrage opportunity can be exploited by:

A) buying the stock QJX, and selling the futures. ✓

B) buying the futures and buying the stock QJX. ✗

C) selling the stock QJX and buying the futures. ✗

Explanation

The calculated fair value of the futures contract is $\$100 \times (1+0.05)^{0.75}$ = $103.73. The asset is relatively underpriced and the futures contract is overpriced. By buying the stock and selling the futures we can lock in a profit greater than the risk-free rate with no risk.

(Module 31.7, LOS 31.l)

Question #100 of 111

Question ID: 1534580

A stock is priced at 38 and the periodic risk-free rate of interest is 6%. What is the value of a two-period European put option with a strike price of 35 on a share of stock using a binomial model with an up factor of 1.15, a down factor of 0.87 and a risk-neutral up probability of 68%?

A) $0.57. ✓

B) $0.64. ✗

C) $2.58. ✗

Explanation

Two down moves produce a stock price of 38 × 0.87^2 = 28.76 and a put value at the end of two periods of 6.24. An up and a down move, as well as two up moves leave the put option out of the money. You are directly given the probability of up = 0.68. Hence, the down probability = 0.32. The value of the put option is [0.32^2 × 6.24] / 1.06^2 = $0.57.

(Module 31.1, LOS 31.b)

Mark Washington, CFA, is an analyst with BIC, a Bermuda-based investment company that does business primarily in the U.S. and Canada. BIC has approximately $200 million of assets under management, the bulk of which is invested in U.S. equities. BIC has outperformed its target benchmark for eight of the past ten years, and has consistently been in the top quartile of performance when compared with its peer investment companies. Washington is a part of the Liability Management group that is responsible for hedging the equity portfolios under management. The Liability Management group has been authorized to use calls or puts on the underlying equities in the portfolio when appropriate, in order to minimize their exposure to market volatility. They also may utilize an options strategy in order to generate additional returns.

One year ago, BIC analysts predicted that the U.S. equity market would most likely experience a slight downturn due to inflationary pressures. The analysts forecast a decrease in equity values of between 3 to 5% over the upcoming year and one-half. Based upon that prediction, the Liability Management group was instructed to utilize calls and puts to construct a delta-neutral portfolio. Washington immediately established option positions that he believed would hedge the underlying portfolio against the impending market decline.

As predicted, the U.S. equity markets did indeed experience a downturn of approximately 4% over a twelve-month period. However, portfolio performance for BIC during those twelve months was disappointing. The performance of the BIC portfolio lagged that of its peer group by nearly 10%. Upper management believes that a major factor in the portfolio's underperformance was the option strategy utilized by Washington and the Liability Management group. Management has decided that the Liability Management group did not properly execute a delta-neutral strategy. Washington and his group have been told to review their options strategy to determine why the hedged portfolio did not perform as expected. Washington has decided to undertake a review of the most basic option concepts, and explore such elementary topics as option valuation, an option's delta, and the expected performance of options under varying scenarios. He is going to examine all facets of a delta-neutral portfolio: how to construct one, how to determine the expected results, and when to use one. Management has given Washington and his group one week to immerse themselves in options theory, review the basic concepts, and then to present their findings as to why the portfolio did not perform as expected.

Question #101 - 104 of 111

Question ID: 1586316

Which of the following *best* explains a delta-neutral portfolio? A delta-neutral portfolio is perfectly hedged against:

A) small price decreases in the underlying asset.

B) all price changes in the underlying asset.

C) small price changes in the underlying asset.

Explanation

A delta-neutral portfolio is perfectly hedged against small price changes in the underlying asset. This is true both for price increases and decreases. That is, the portfolio value will not change significantly if the asset price changes by a small amount. However, large changes in the underlying will cause the hedge to become imperfect. This means that overall portfolio value can change by a significant amount if the price change in the underlying asset is large.

(Module 31.7, LOS 31.l)

Question #102 - 104 of 111

Question ID: 1586317

After discussing the concept of a delta-neutral portfolio, Washington determines that he needs to further explain the concept of delta. Washington draws the payoff diagram for an option as a function of the underlying stock price. Using this diagram, how is delta interpreted? Delta is the:

A) slope in the option price diagram.

B) level in the option price diagram.

C) curvature of the option price graph.

Explanation

Delta is the change in the option price for a given instantaneous change in the stock price. The change is equal to the slope of the option price diagram.

(Module 31.7, LOS 31.l)

Question #103 - 104 of 111

Question ID: 1586318

Washington is trying to determine the value of a call option. When the slope of the at expiration curve is close to zero, the call option is:

A) in-the-money.

B) at-the-money.

C) out-of-the-money.

Explanation

When a call option is deep out-of-the-money, the slope of the at expiration curve is close to zero, which means the delta will be close to zero.

(Module 31.7, LOS 31.l)

Question #104 - 104 of 111

Question ID: 1586319

BIC owns 51,750 shares of Smith & Oates. The shares are currently priced at $69. A call option on Smith & Oates with a strike price of $70 is selling at $3.50, and has a delta of 0.69 What is the number of call options necessary to create a delta-neutral hedge?

A) 0.

B) 75,000.

C) 14,785.

Explanation

The number of call options necessary to delta hedge is = 51,750 / 0.69 = 75,000 options or 750 option contracts, each covering 100 shares. Since these are call options, the options should be sold short.

(Module 31.7, LOS 31.l)

Question #105 of 111

Question ID: 1473740

To the issuer of a floating rate note, a cap is equivalent to:

A) writing a series of interest rate calls.

B) owning a series of interest rate calls.

C) owning a series of calls on a fixed income security.

Explanation

The issuer of the note is borrowing at a floating rate, and will have higher interest expenses if rates increase. A cap is equivalent to owning a series of interest rate calls at the cap rate that will pay the difference between the market rate and the cap rate. If interest rates increase, the payoff from the calls will compensate the borrower for the higher interest expenses.

(Module 31.6, LOS 31.j)

Question #106 of 111

Question ID: 1473706

A bond analyst decides to use the BSM model to price options on bond prices. This model will *most likely* be inadequate because:

A) BSM cannot be modified to deal with cash flows like coupon payments. ✗

B) the price of the underlying asset follows a lognormal distribution. ✗

C) the risk free rate must be constant and known. ✓

Explanation

The BSM model is not useful for pricing options on bond prices and interest rates. In those cases, interest rate volatility is a key factor in determining the value of the option. BSM can be modified to deal with cash flows like coupon payments. The assumption that "the price of the underlying asset follows a lognormal distribution" is not applicable.

(Module 31.6, LOS 31.f)

Question #107 of 111

Question ID: 1473736

Cal Smart wrote a 90-day receiver swaption on a 1-year MRR-based semiannual-pay $10 million swap with an exercise rate of 3.8%. At expiration, the market rate and MRR yield curve are:

Fixed rate 3.763%

180-days 3.6%

360-days 3.8%

The payoff to the writer of the receiver swaption at expiration is *closest* to:

A) $0. ✗

B) -$3,600. ✓

C) \$3,600.

Explanation

At expiration, the fixed rate is 3.763% which is below the exercise rate of 3.8%. The purchaser of the receiver swaption will exercise the option which allows them to receive a fixed rate of 3.8% from the writer of the option and pay the current rate of 3.763%.

The equivalent of two payments of (0.038 - 0.03763) × (180/360) × (10,000,000) will be made to the receiver swaption. One payment would have been received in 6 months and will be discounted back to the present at the 6-month rate. One payment would have been received in 12 months and will be discounted back to the present at the 12-month rate

The first payment, discounted to the present is (0.038 - 0.03763) × (180/360) × (10,000,000) × (1/1.018) = \$1,817.28.

The second payment, discounted to the present is (0.038 - 0.03763) × (180/360) × (10,000,000) × (1/1.038) = \$1,782.27

The total payoff for the writer is -\$3,599.55.

(Module 31.6, LOS 31.j)

Question #108 of 111

Question ID: 1473744

Which of the following option sensitivities measures the change in the price of the option with respect to a decrease in the time to expiration?

A) Gamma.

B) Delta.

C) Theta.

Explanation

Theta describes the change in option price in response to the passage of time. Since option holders would prefer that value not decay too quickly, an option with a low theta value is desirable.

(Module 31.7, LOS 31.k)

Question #109 of 111

Question ID: 1473738

Which of the following *best* describes an interest rate cap? An interest rate cap is a package or portfolio of interest rate options that provide a positive payoff to the buyer if the:

A) T-Bond futures exceeds the strike price.

B) reference rate exceeds the strike rate.

C) reference rate is below the strike rate.

Explanation

An interest rate cap is a package of European-type call options (called caplets) on a reference interest rate.

(Module 31.6, LOS 31.j)

Question #110 of 111

Question ID: 1473754

The price of a June call option with an exercise price of $50 falls by $0.50 when the underlying non-dividend paying stock price falls by $2.00. The delta of a June put option with an exercise price of $50 *closest* to:

A) –0.25.

B) 0.25.

C) –0.75.

Explanation

The call option delta is:

$$\text{delta}_{\text{call}} = \frac{\$0.50}{\$2.00} = 0.25$$

The put option delta is 0.25 – 1 = –0.75.

(Module 31.7, LOS 31.l)

Question #111 of 111

Question ID: 1473747

The delta of an option is equal to the:

A) dollar change in the stock price divided by the dollar change in the option price.

B) dollar change in the option price divided by the dollar change in the stock price.

C) percentage change in option price divided by the percentage change in the asset price.

Explanation

The delta of an option is the dollar change in option price per $1 change in the price of the underlying asset.

(Module 31.7, LOS 31.k)